

		Centripetal force = $m\frac{v^2}{r} = \frac{2m\pi r}{r} = 2m\pi g$ $2mg = tension - mg \rightarrow tension = 3mg$
10	D	$F = -\frac{dU}{dr}$, hence option D is correct while option A and B are incorrect. For option C, since the $F-r$ graph is for the small point mass, area under the $F-r$ graph will only provide the U of the small point mass. (U is the work done by external agent in bringing small point mass from infinity to that that point.)